

CONTEXT 3

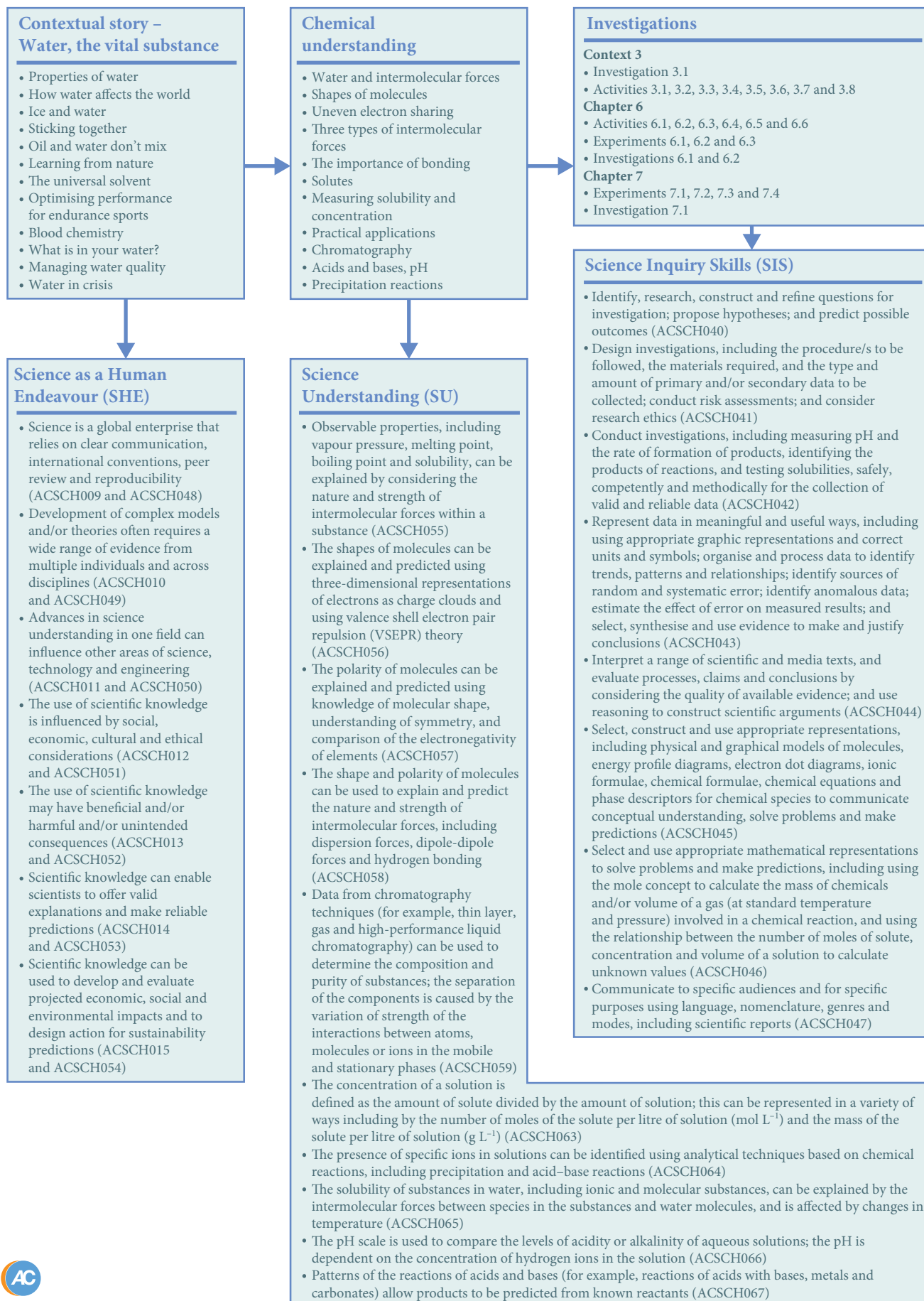
WATER, THE VITAL SUBSTANCE

By the end of this chapter you will have covered the following material.

Science as a Human Endeavour

- Science is a global enterprise that relies on clear communication, international conventions, peer review and reproducibility ([ACSCH009](#) and [ACSCH048](#))
- Development of complex models and/or theories often requires a wide range of evidence from multiple individuals and across disciplines ([ACSCH010](#) and [ACSCH049](#))
- Advances in science understanding in one field can influence other areas of science, technology and engineering ([ACSCH011](#) and [ACSCH050](#))
- The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations ([ACSCH012](#) and [ACSCH051](#))
- The use of scientific knowledge may have beneficial and/or harmful and/or unintended consequences ([ACSCH013](#) and [ACSCH052](#))
- Scientific knowledge can enable scientists to offer valid explanations and make reliable predictions ([ACSCH014](#) and [ACSCH053](#))
- Scientific knowledge can be used to develop and evaluate projected economic, social and environmental impacts and to design action for sustainability ([ACSCH015](#) and [ACSCH054](#))





Water is a tasteless, odourless liquid, which is essential to our survival. All organisms are at least 60% water. Although water is very common, it has particular properties that make it an unusual material. These properties have enabled life to evolve on Earth and explain why water is vital for life.

The importance of water was recognised by Empedocles (490–430 BCE). He claimed that water was one of the four elements that formed everything that we know. But he never realised the reasons why this substance is so vital for life.

Water has been found not only on Earth but also in black holes, on Mars and on Saturn's moon Titan. But life is only expected to occur where water exists as a liquid.



GOLDILOCKS PLANETS

Visit this website to learn about NASA's search for habitable planets.



Dreamstime/Antartix

◀ Figure 3.1

Earth is a Goldilocks planet. It is not too hot and it is not too cold. Just perfect: it can have liquid water.

WOW

The hunt for Goldilocks planets

Venus is too hot, Mars is too cold and Earth is just right. NASA's astronomers are looking for planets around other stars in the Milky Way that are also just right. The planets have to be in the habitable zone of a solar system – where temperatures allow water to exist as a liquid. The astronomers are looking for planets that may have liquid water. Evidence of planets in other solar systems has only recently been found. Now, hundreds of planets have been found, including a few Earth-size planets. It is these planets that may contain life.



To learn about how molecular shape is determined and how the shape influences the bonding between molecules, refer to Chemistry section 6.2 on page 258.

Development of complex models and/or theories often requires a wide range of evidence from multiple individuals and across disciplines (ACSCH010 and ACSCH049)



Properties of water

Water freezes at 0°C and boils at 100°C. It can act as a **solvent** over this entire temperature range, dissolving and transporting materials across a cell or over the oceans. It is the only substance on Earth that exists naturally in its three states of matter: solid, liquid and gas. Unlike other substances, solid water is less dense than the liquid. So ice floats.

The boiling point, density in solid and liquid phases, **surface tension** and its ability to act as a solvent are the unique properties of water. These properties are due to the water molecule's shape, bonding within the molecule and the bonds it forms with other substances. Ultimately it is water's bonding that makes it vital in biological, chemical and physical processes on Earth.

How water affects the world

Figure 3.2 ►

The tip of the iceberg:
why do icebergs float in
water?



Scientists study ocean currents when they are investigating the climate and marine environments. Cold, salty water is denser than warm, fresh water. Ocean currents are generated when cold, salty surface water sinks and moves by convection towards the equator along the ocean floor. The less dense warmer water flows in from nearby regions along the surface.

Sailors in the tropics first noticed the stream of freezing cold water near the bottom of the ocean. This cold water must have come from the Arctic. This cold stream served the useful purpose of chilling bottles of wine dunked overboard.

The currents may take

more than 500 years to move over the planet but they have a dramatic effect on our climate. Oceanographers, meteorologists, software engineers, mathematicians and chemists have developed complex climate models that are used to predict the effects of events such as **El Niño** on Australia's climate.

The El Niño events highlight the link between water and Australia's climate. During an El Niño event, there is a warmer than usual ocean current off the South American coast. This leads to cooler sea surface temperatures across the Pacific. We feel this along the east coast, with drier conditions, more droughts and more severe bushfire seasons. Paradoxically, warmer tropical winds result in more floods in the north of Australia.

The temperature and salt levels of the oceans both affect the currents. A sample of Antarctic water would weigh more than an equal volume of fresh, warm water. Antarctic water is saltier partly because of sea ice. As sea ice forms, dissolved salt is squeezed out. The salt cannot be part of the ice crystal. The expelled salt remains dissolved in the surrounding water. This helps make Antarctic water the densest, saltiest water on the planet.

At the surface, carbon dioxide and oxygen gases dissolve into the ocean. They dissolve better in cold water. Cold water sinks, taking oxygen and carbon dioxide to great depths. This provides a sink for carbon dioxide from our atmosphere and a source of oxygen for the life at these depths. This explains why there is so much life in Antarctic oceans.

Oceans affect the water cycle. The ocean is the main source of the water that is evaporated, cooled, condensed and precipitated. Flowing rivers cause weathering of rocks, washing sediments and dissolved salts into oceans, lakes and underground aquifers. During this cycle, water will go from gas to liquid to solid. Solid water forms the ice that breaks rock cracks open, falls as snow or forms sea ice in the Arctic and Antarctic. The unique properties of water affect these processes and have a dramatic effect on life on Earth.



OCEAN CURRENTS 1

Visit this website to learn more about ocean currents and how this knowledge is supporting our understanding of climate change.



CSIRO Australia

Figure 3.3 ▲
Water currents around Australia



OCEAN CURRENTS 2

Visit this website to learn how ocean currents are affecting the Australian climate and marine environment.

INVESTIGATION 3.1

DENSITY CHANGES IN WATER DUE TO TEMPERATURE AND SALINITY

Two factors affect the density of water: temperature and salinity (the concentration of salt). Density is a measure of mass (g) per unit volume. Ocean currents are formed as the cold salty water sinks underneath the warmer fresher water.

Aim

To design an experiment to determine which factor, temperature or salinity, has the greater effect on the density of water.

You will need

- Accurate thermometer (digital thermometer) that measures from 0°C to 50°C
- Glass measuring cylinder
- Electronic balance
- Source of heating or cooling
- Distilled water
- Salt

You can prepare a range of samples by adding 50 g of table salt to 500 mL of distilled water. This will make the solution equivalent to 100 g of salt per litre of solution. Suggested dilutions are:

- Add 50 mL of distilled water to 50 mL of the solution to make a 50 g L⁻¹ sample.
- Add 80 mL of distilled water to 20 mL of the solution to make a 20 g L⁻¹ sample.
- Add 90 mL of distilled water to 10 mL of the solution to make a 10 g L⁻¹ sample.

What are the risks?

Construct a table similar to the one below. Identify specific risks involved in the investigation and ways that you will manage the risks to avoid injuries or damage to equipment. Ask your teacher to check your risk assessment before you proceed.

What are the risks in doing this investigation?	How can you manage these risks to stay safe?

How will you carry out your investigation?

- 1 How will you measure the density of each solution?
- 2 How will you heat or cool your solution?
- 3 Will you start with a warm solution and cool it down? Or will you prepare fresh solutions each time? Or will you start with a cool solution and warm it up?

How reproducible was your method?

- 1 What results will you collect?
- 2 How many samples will you test?
- 3 How many temperatures will you choose?
- 4 What range of temperatures will you choose?
- 5 Will you test fresh water?
- 6 Collate your results in a table similar to the following and give your table a title.

Salinity	Temperature 1	Temperature 2	Temperature 3

How will you analyse your results?

Will you graph the results? What type of graph will you use: a column graph or a line graph?

What have you found?

- 1 How does the temperature affect the density of water?
- 2 How does the salinity affect the density of water?

What do you conclude?

- 1 From your results, which factor had the biggest effect on the density?
- 2 Explain how the density changes with temperature and solute levels.

Ideas for improvement

From your results, which method gave the most consistent measure of density?

Taking it further

Some soft drinks contain a large amount of sugar or other additives. Apply your method to determine the effect of sugar levels on density. Was it a similar to result salt levels?

Further extension

How is this knowledge used in the fruit juice, wine, brewing, sugar and honey industries?

Ice and water

Water is the only substance that expands on freezing. As water cools, the molecules become closer together, so the density increases until the temperature approaches 4°C . After this point, the water begins to expand and become less dense. Ice floats. For all other substances, the solid form is denser than the liquid and sinks.

Ice has a definite hexagonal pattern of water molecules. This arises through the interaction between the bonding within and between the water molecules. The molecules are further apart in ice than in liquid water, which is why ice is less dense than liquid water.

If ice did sink, then in the cold regions of the world and during the Ice Ages, the bottom of the lakes and oceans would remain permanently frozen. Never being exposed to the sun, the layer of ice would increase until the whole lake or ocean froze solid. This would kill all life in the oceans and lakes. However, a floating layer of ice insulates the water beneath from freezing. Animals and plants can exist in lakes when the air temperature is below 0°C .

It is difficult to live in near frozen conditions because of the threat of water in cells freezing. With organisms composed mostly of water, the formation of ice can be lethal because an ice crystal can puncture the cell membrane. Fresh water freezes at 0°C and salty water will freeze at -2°C . Frostbite occurs when ice crystals are formed in and around skin cells when a person is exposed to extreme cold. However, life does exist under extremely cold conditions. One reason is that some organisms' cells can produce antifreeze agents. These chemicals, typically salts or proteins, disrupt the ice crystal structure. Some plant cells do not freeze until -40°C . This enables those plants to survive in frozen areas through the winter.

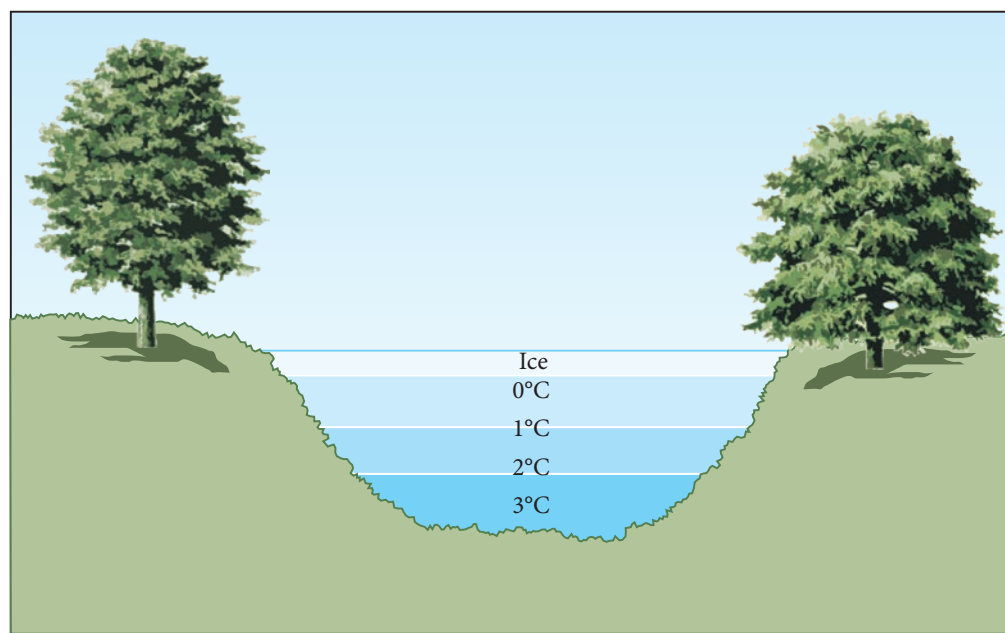
Frozen blood and blood products are used by the armed forces during conflict when haemorrhage is a major cause of death of soldiers. There are limited options for treating blood loss on the battlefield and logistically it is difficult to rely on fresh blood. Frozen blood and products can be stored for years at -80°C . The blood products are not snap-frozen but treated with glycerol and other chemicals to stop ice crystals forming. Within 90 minutes, which includes 30 minutes of thawing, red blood can be treated and ready to use.

By understanding the chemistry that is occurring in the blood, the army has been able to integrate fresh and frozen blood products to save lives on the battlefield. This system could also be used in isolated rural areas of Australia. This is an example of how the understandings developed by chemists are applied in other areas of science, in this case biology and medicine.

Advances in science understanding in one field can influence other areas of science, technology and engineering (ACSCH011 and ACSCH050)



To understand more about why ice has a lower density than water, refer to Chemistry section 6.5 on 269.



◀ **Figure 3.4**

Life under the ice. Ice is less dense than liquid water so floats on top. Colder water is less dense than warmer water so sits above it. The layer of ice insulates the water below it against freezing.

ACTIVITY 3.1

MAKING ICE CREAM

The effect on freezing point of the interaction between solutes and water molecules is not always detrimental. Research on the Internet to find out how ice-cream manufacturers use the effect of solutes on the melting point of water to make ice cream. A useful search term would be 'freezing point depression'.

The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations (ACSCH013 and ACSCH052)



IVF: frozen embryos or frozen eggs?

The human egg (oocyte) is the single largest cell in the body. It is harder to freeze than an embryo. During in vitro fertilisation (IVF) procedures, embryos are frozen rather than the woman's eggs. If the egg were placed in a freezer, ice crystals would rupture the cell membrane. To avoid this, the water has to be removed from the egg and replaced with a **cryoprotectant** ('antifreeze solution').

Research is being undertaken to successfully freeze eggs. This will allow women to freeze their eggs before treatments such as chemotherapy, which would normally destroy their eggs. Improving the technique may remove some ethical concerns that some people have about frozen embryos. These concerns include what to do with excess embryos that result from IVF treatment, especially if the donor couple cannot decide or can no longer be contacted. Excess embryos can be frozen for later treatments, donated for medical research or to infertile couples, or allowed to thaw and be destroyed. Some couples will only donate the embryo if they can determine the age, race, socioeconomic status or religion of the infertile couple. If the original couple separate, or the man dies, then the status of the embryo is not certain. The legal system is struggling to keep up with the advances in technology.

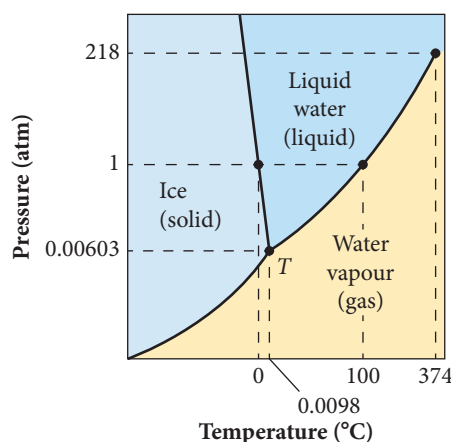
Freezing eggs also has ethical issues. Fertility drops dramatically after 35 years of age, so some women want to freeze their eggs until they are ready to have children. Is this a medical or a social issue? Women undergoing IVF treatment also need to be aware that the chances of having a healthy carefree pregnancy also decrease with age. Also, the effects of the cryoprotectants on the chromosomes, development and survival rate of the embryo are not completely known.

WOW

Surprise

Scientists are still being surprised by nature. Lake Vostok is the largest subglacial lake in Antarctica and the seventh largest lake in the world. It was first identified in 1950 beneath the Russian Vostok Station, but its existence would not be confirmed until the 1990s. It is a freshwater lake. On 10 January 2013, Russian scientists successfully obtained a core of freshly frozen water from the lake. RNA (ribonucleic acid) was found in the core, which has led to questions about the type of organisms in the lake, which has not been in contact with the atmosphere for millions of years.

Temperatures in Antarctica can get as low as -89°C . Lake Vostok is 500 metres below sea level and the average temperature of the water is -3°C , well below the freezing point of water. It has about 4 km of ice above it, yet it contains liquid water. A similar phenomenon is observed when water is formed under a glacier and under the blades of ice skaters.



▲ Figure 3.5

The state of water depends upon its temperature and pressure, as shown in this triple point (T) graph.

Perfecting the temperature

Water affects all biology on Earth through its effect on heat exchange. The cooling breeze off an ocean can be very welcome. In Perth, the breeze that cools the city after a hot summer day is called the Fremantle Doctor. Water has a high specific heat capacity so it takes a lot of energy or heat to warm the ocean. Large bodies of water have a more constant temperature than land.

The body's reactions are catalysed by proteins called **enzymes**, which operate optimally at about 36°C. If the body gets too hot, then the reactions will stop. Similarly, if the body gets too cold, then the reactions are too slow to sustain life. Water's high specific heat means that cell temperatures usually remain within the narrow range required for life.

One function of blood is to regulate body temperature. As you exercise, blood flow increases and you sweat. Energy is absorbed from the environment when a liquid changes state to a gas. You feel cool as the water in sweat absorbs heat and changes state.

Sticking together

Water molecules stick together. The high surface tension explains the **viscosity**, or stickiness, of water. It may appear very slight, but this high surface tension, which allows insects to walk on water, has provided the evolutionary pressure for selection of a streamlined shape, and affects the blood and fluids flowing through your body.

Surface tension explains why oil and water do not mix. Water's surface tension is high and the oil molecules cannot break the molecules apart to mix with it.

Oil and water don't mix

In 1989, the Exxon Valdez super tanker crashed into Prince William Sound, Alaska, USA. More than 40 000 tonnes of oil spilt in the pristine environment. The oil spill contaminated over 2000 km of coastline, killing record numbers of sea life: tens of thousands of sea birds, thousands of sea otters, hundreds of seals and more than 20 orca whales. Despite US\$2.5 billion



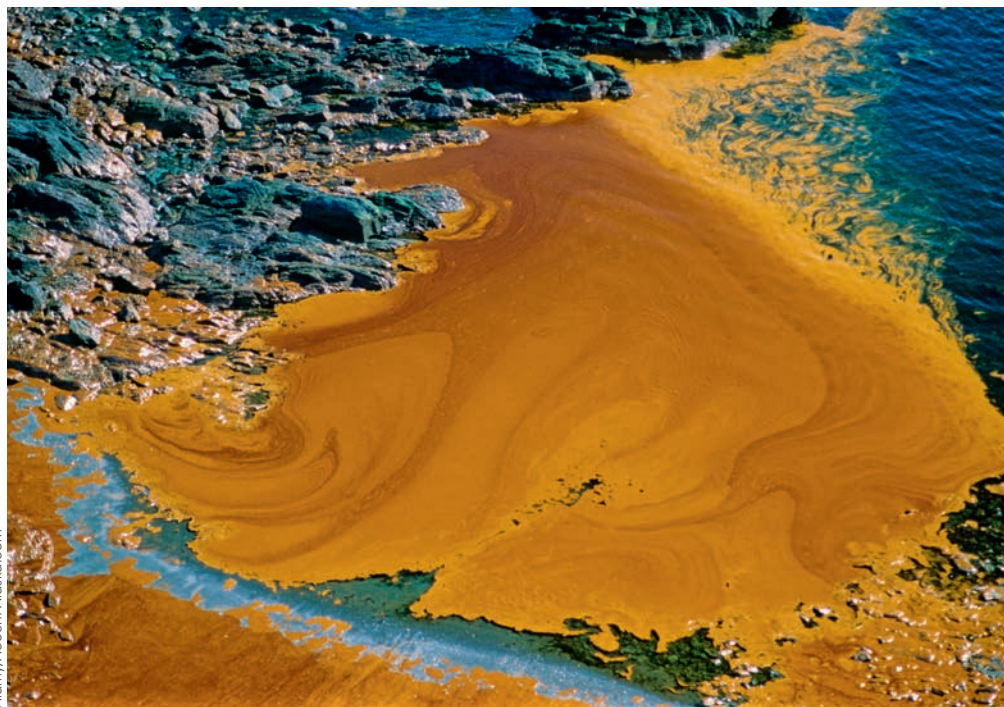
To revise specific heat of capacity and latent heat, refer to Chemistry section 4.3 on page 213.



To understand more about the intermolecular bonds and surface tension in water, refer to Chemistry sections 6.5 and 6.8 on pages 269 and 286.

The effect of extremes of temperature on enzymes will be discussed in Context 4 'Making reactions work for us'.

Advances in science understanding in one field can influence other areas of science, technology and engineering (ACSCH011) and ACSCH050



Alamy/Accent Alaska.com

▲ Figure 3.6

The Exxon Valdez oil spill caused damage to the environment in Prince William Sound, Alaska, USA.



SURFACE TENSION IN SPACE

Watch this video to see how the surface tension of water is affected by gravity.



To understand more about solubility, refer to Chemistry section 6.6 on page 275.

The Exxon Valdez accident was depicted in an episode of the Simpsons, Bart after Dark, in which Lisa cleaned the rocks of oil. The ditty 'What do you do with a drunken sailor? Put him in charge of an Exxon tanker' highlighted one of the main causes of the disaster – the captain was drunk.



To learn more about surfactants and chromatography, refer to Chemistry section 6.8 on page 286.



NANOTECHNOLOGY

Watch this video of demonstrations of the benefits of nanotechnology and some amazing advances that are just around the corner.

being spent on the clean-up, oil still lingers on the coast today. The wild life, such as sea otters and fish, are not recovering as well as hoped. The fishing industry collapsed in 1993.

Oil and most organic compounds are **hydrophobic** ('water-hating') and do not dissolve in water. Substances that dissolve easily in water are called **hydrophilic** ('water-loving'). Hydrophobic substances will form a layer either above or below the water. The oil from the disaster has not been washed away with the water but is found under the surface of some beaches near the intertidal zone.

Chemists can confirm that the oil is from the Exxon Valdez disaster and not from a new deposit spill by identifying components of the oil by very sensitive techniques such as **chromatography**, in particular high-performance liquid chromatography.

Mixing it up

Surfactants are molecules that are used to break the surface tension of water. In this way, they can allow oil and water to mix. Surfactants are used in many industries and are a major component of household cleaning agents. Detergents are one type of surfactant that are used to wash away the grease from dirty plates. Surfactants are generally used in oils spills to disperse the oil. In the Exxon Valdez oil spill, weather and tidal factors reduced the effectiveness of the surfactants.

The Exxon Valdez has become the benchmark for other oil spills. Spills are compared to this spill to indicate the relative size of the accident. But it also spurred an industry that aimed to minimise the environmental and economic damage from such incidents. When other disasters like the Gulf of Mexico oil spill occurred in 2010, expertise gained in the Exxon Valdez incident was applied to try to minimise the effect on the local fishing and tourism industry.

Learning from nature

Scientists are mimicking hydrophobic interactions in nature to develop nanomaterials. Nanostructures in the lotus leaf force water into beads so it runs off the leaf. This process is being mimicked in the laboratory to develop self-cleaning surfaces. Nanopatterned surfaces are being developed with nano bumps to generating water-grabbing and water-shedding materials.

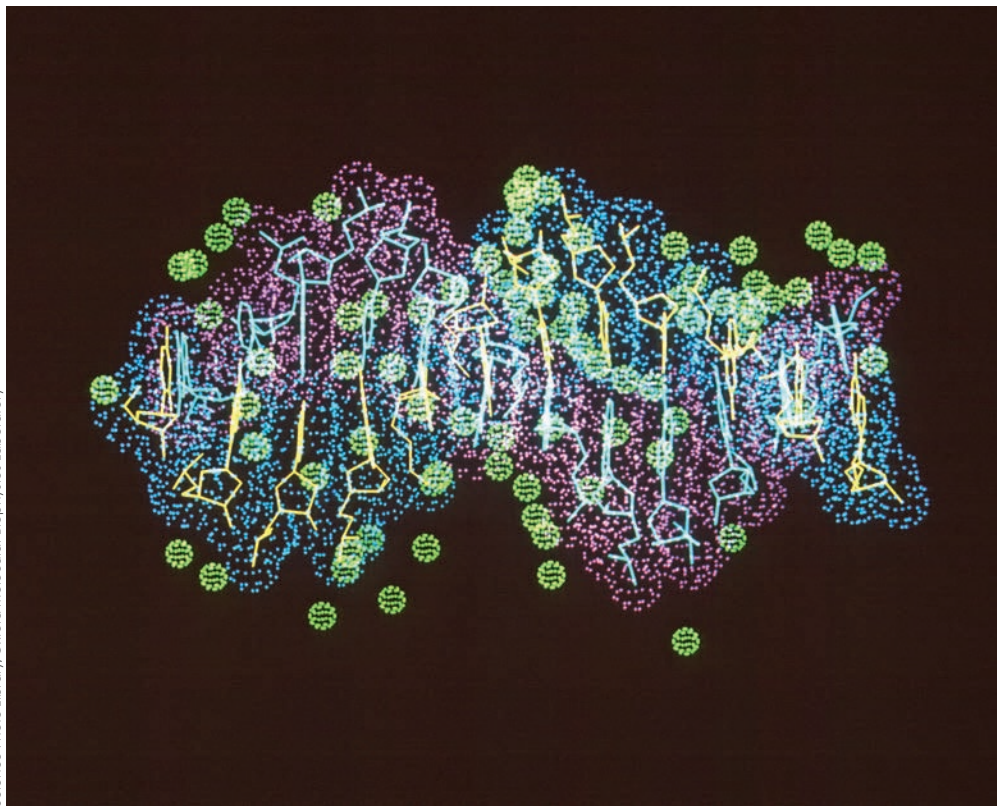
Interactions between oil-type molecules and water are the foundation of life. These interactions allow the formation of cell membranes and the folding of protein and nucleic acids. By copying and manipulating these interactions, scientists have produced nanomaterials that self-assemble.

One current area of research is superhydrophobic surfaces. These could lead to better heat transfers in power and desalination plants.

Water and biomolecules interact

When some organic compounds such as protein, RNA and DNA crystallise, water is included in the crystals. This is called **water of crystallisation**.

To determine the structure of a protein, scientists often need to get it in crystal form. Up to 50% of a protein crystal could be due to water. To make a protein crystal, a **supersaturated solution** (an unstable **solution** that has additional **solute** dissolved for the particular temperature) of the protein is made, and then the crystal is allowed to slowly precipitate out. It is important that the protein remains folded while the structure is determined. To do this scientists have to control the water content consistently.



◀ **Figure 3.7**

DNA and water interact. A DNA molecule has a hydrophilic backbone and a hydrophobic inner region of nitrogenous bases (adenine, guanine, thymine and cytosine). The hydrophobic region repels any water (green) between them, which forces the bases towards each other.



To learn about supersaturated solutions and water of crystallisation, refer to Chemistry sections 6.5 and 6.7 on pages 269 and 279.

WOW

The different shapes of DNA crystals when water is present

Rosalind Franklin (1920–58) used her understanding of chemistry and her skills in X-ray diffraction to produce a superior image of the DNA molecule. Under controversial circumstances, her co-worker Maurice Wilkins showed the image to James Watson. This was the proof needed to confirm that DNA was a helix. Franklin's analysis had also enabled her to point out errors in Watson and Crick's earlier models.

In 1962, the Nobel Prize for Physiology or Medicine was awarded to Watson, Crick and Wilkins for the discovery of the structure of DNA. Did Franklin miss out because Nobel Prizes are not awarded posthumously? Or was it due to the sexist attitude of the 1940s and 1950s? Franklin had attended one of the only schools that taught Chemistry and Physics to girls. Her work has since been acknowledged.

Development of complex models and/or theories often requires a wide range of evidence from multiple individuals and across disciplines (ACSCH010 and ACSCH049)



In Unit 4 Chemistry, you will examine how chemists change one molecule into another.

The organic functional groups $-OH$, $-C=O$ and $-NH_2$ will be covered in detail in Unit 3.

Manipulating molecules

The shape and structure of molecules will affect how they behave. The shape and structure affects the intermolecular bonds that are formed. By changing or adding different groups to the molecule, the properties of the material changes.

Organic molecules tend to be hydrophobic because they are mostly non-polar. Adding polar groups such as the hydroxyl (OH) group may make the molecule become polar. Polar molecules are hydrophilic.

Methane (CH_4) and ethane (CH_3CH_3) are hydrophobic and form dispersion forces with water. Methanol (CH_3OH) and ethanol (CH_3CH_2OH) can form hydrogen bonds with water and will dissolve easily in water.

ACTIVITY 3.2

VITAMINS: GOOD OR BAD?

Vitamins are a group of essential nutrients that we only need in small amounts. They are a diverse group of organic compounds. For example, unlike other animals, humans must consume vitamin C. A lack of vitamin C causes scurvy – a deadly disease that used to kill sailors on long voyages. In 1770, Captain James Cook accepted the scientific view that a diet including citrus fruit was essential. He was the first to circumnavigate the world without losing a man to scurvy. An excess of vitamin C is never fatal.

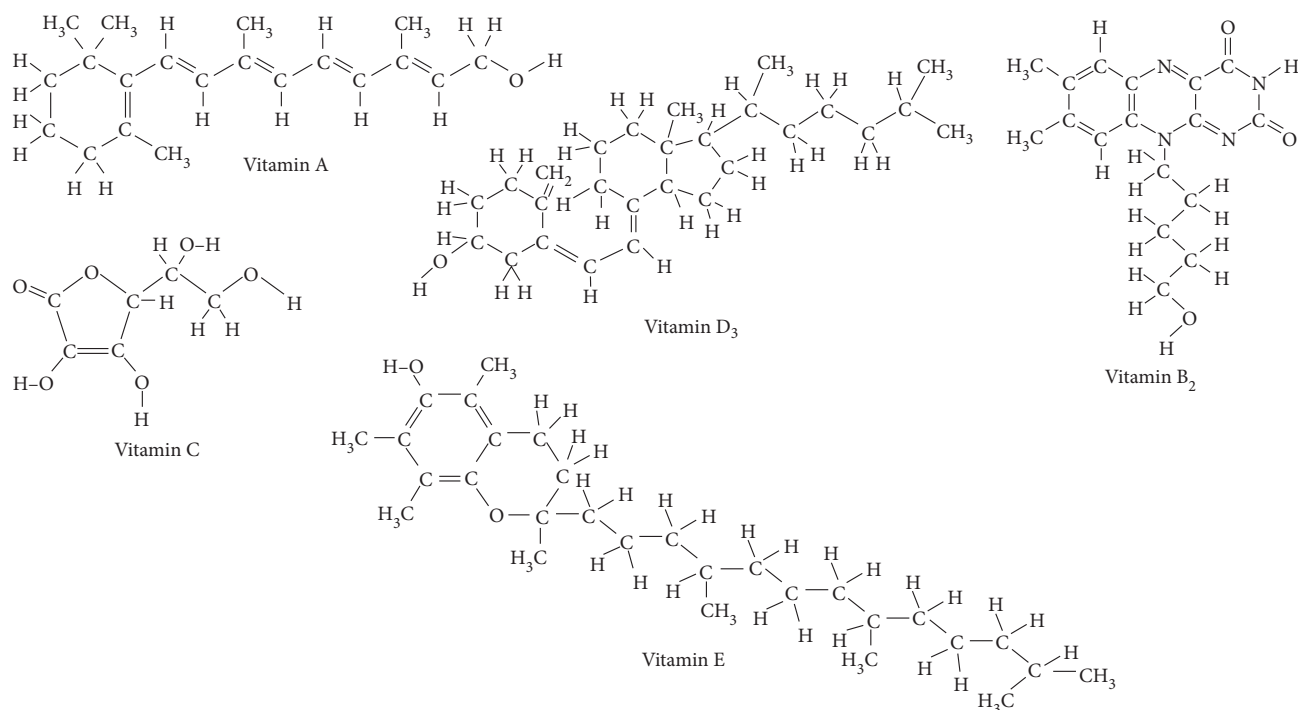
Vitamin A is believed to have caused the death of Antarctic explorer Xavier Mertz in 1913. Short of food Mertz and fellow explorer Douglas Mawson resorted to eating their sledge dogs. Dog livers are high in vitamin A and an excess of this vitamin could have caused his death.

Vitamin A is a non-polar fat-soluble vitamin. It accumulates in the liver and is harder to remove from the body than water-soluble vitamins. This leads to it being toxic in large doses.

- Copy Figure 3.8. For each vitamin, identify and circle the polar groups.
- Why is vitamin C considered polar and water-soluble?
 - Why is vitamin A considered non-polar and fat-soluble?
 - How does this explain why an excess of vitamin C is never fatal but excessive vitamin A can be fatal?
- Decide whether the other vitamin molecules overall are polar or non-polar. Justify your decisions.
- On the basis of polarity, classify vitamins B, D and E as likely to be water- or fat-soluble. Research to see if your predictions are correct.

Extension

Research how excess chemicals are removed from the body.



▲ **Figure 3.8**
Vitamins A, B, C,
D and E

Surfactants and premature babies

Babies born prematurely often die when they struggle for breath. In the 1950s, US paediatrician Dr Mary Ellen was puzzled as to why there was no residual air in the lungs of these babies. It appeared that their lungs could not retain air and collapsed every time the baby breathed out.

Dr Ellen and others that followed discovered the reason behind one of the major causes of death in preterm babies. If babies are born too early they are unable to produce a lung surfactant. The surface water tension in lungs is so high that the alveoli (air sacs) collapse. The treatment is to supply artificial surfactant to the baby. The surfactant breaks the surface tension, allowing the baby to take in a breath of air. This will occur until the baby is able to make its own surfactant, typically in a few days' time. With enough warning, the mother can also be given steroids that stimulate the baby's lungs to start producing the surfactant. It took until 1980 before the first human babies started to benefit from the use of surfactants.

Around 1989, another treatment started to be trialled – the use of 'liquid breathing'. When babies are born very prematurely, doctors fill their lungs with perfluorocarbons (PFCs). Perfluorocarbons are hydrocarbons in which the hydrogen atoms have all been replaced with fluorine atoms. Perflubron ($C_8F_{17}Br$) (Figure 3.10) is also included in this class of compounds, even though it contains bromine. Perflubron is an excellent medium for carrying respiratory gases. These liquids can carry large amounts of oxygen and carbon dioxide and reduce the surface tension, allowing the baby to breathe.

It was a strange birth for this range of molecules. During the development of the atom bomb (The Manhattan Project 1942–5), chemists tried to develop an inert (unreactive) solvent for the highly reactive uranium chemicals. What they produced was perfluorocarbons. These remarkable liquids are clear, odourless and very dense and can carry 20 times as much oxygen as other fluids. Perfluorocarbons are now also being used as blood substitutes.



Corbis/J.P. Laffont/Sygma

The use of scientific knowledge may have beneficial and/or harmful and/or unintended consequences (ACSCH013 and ACSCH052)



LIQUID BREATHING

Watch this video to see a mouse breathe in liquid, and some future applications.

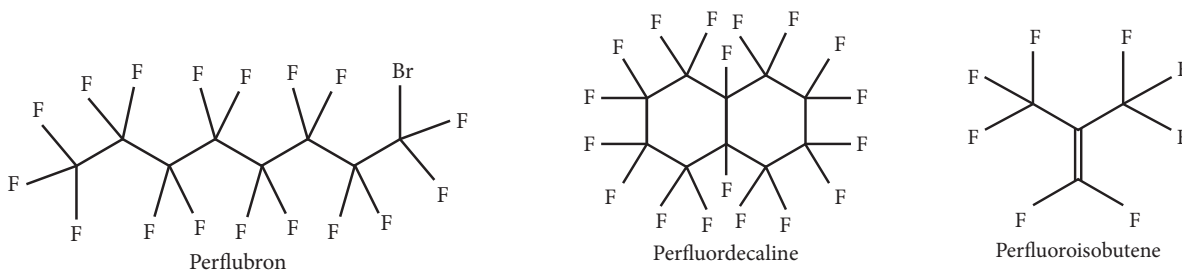
◀ Figure 3.9

This mouse is breathing in a liquid-filled jar, demonstrating liquid breathing.

ACTIVITY 3.3

PERFLUOROCARBONS

The structure of a perfluorocarbon molecule gives remarkable functionality to the material.



▲ Figure 3.10 A range of perfluorocarbon molecules

- What type of bonding would you expect between the molecules of a perfluorocarbon?
- Explain why perfluorocarbons are essentially insoluble in water and alcohols and only sparingly soluble in some lipids.
- Would you expect the surface tension of perfluorocarbons to be high or low? Explain your reasoning.

Extension

The solubility of gases in the liquid perfluorocarbon is due to the presence of intermolecular cavities. Explain how this is different from and similar to how gases are dissolved in water.

The universal solvent

Three-quarters of the planet is covered with water, which exists as a liquid between 0°C and 100°C. Water dissolves both organic materials, such as enzymes and amino acids, and inorganic materials, such as salts and minerals. Generally, water has a low viscosity, remaining a liquid rather than becoming a thick slimy soup. Oxygen has to be dissolved in water for our cells to access it. Water plays a key role in the body's hydrolysis reactions in which it is split into H^+ and OH^- ions, which you will probably recognise as the ions involved in acid and base reactions. When carbon dioxide reacts with water, it forms a weak acid, releasing H^+ ions, which play a role in maintaining the cell's environment. Later you will see how rain that contains H^+ ions can dissolve rocks. A substance's solubility in water is determined by intermolecular forces.

Water's unique ability to act as a universal solvent can explain the chemical processes on Earth and the biological processes in our bodies.

When a substance mixes in water, it may no longer be visible. But it doesn't disappear. The substance may have either reacted with or dissolved in the water. Substances dissolving in water affect the solvent by changing its density, and melting and boiling points.

Soluble salts and soluble organic substances

The water in the ocean has a lot more material or solute dissolved in it than fresh water. You can tell this by the taste or because you can float easier in the ocean.

Seawater typically contains six major ions: sodium (Na^+), chloride (Cl^-), sulfate (SO_4^{2-}), magnesium (Mg^{2+}), calcium (Ca^{2+}) and potassium (K^+). River water has lower levels of all salts and a higher proportion of calcium (Ca^{2+}) and hydrogen carbonate (HCO_3^-). Obviously you can't simply get seawater from evaporating river water. The salt levels present are the result of weathering of mostly igneous rocks, erosion, land run-off and the precipitation of different ions onto the floor of the ocean.

Every naturally occurring element is also present in the ocean. Unfortunately, because of pollution, some are at a higher than desired level.

The main ions in your body fluids are sodium (Na^+), potassium (K^+), chloride (Cl^-), calcium (Ca^{2+}), magnesium (Mg^{2+}), bicarbonate (HCO_3^-), phosphate (PO_4^{2-}) and sulfate (SO_4^{2-}). This is very similar to seawater.

In Chemistry Chapter 7, you will learn more about the role of H^+ and OH^- in acid and bases. Acid and bases have key roles in many of the biochemical reactions.



To learn more about why materials may or may not dissolve in water and these reactions, refer to Chemistry section 6.6 on page 275.

The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations (ACSCH012 and ACSCH051)



WOW

Miracle waters

Mineral waters have been thought to have curative properties for over 2000 years. The common minerals in mineral water are Mg^{2+} , Ca^{2+} and K^+ . Mineral waters can also include small amounts of arsenic, cadmium, chromium and cyanide. Some mineral waters even have abnormally high levels of radioactivity owing to the presence of radioactive elements such as uranium, radon and thorium. At one time, rather than a cause for alarm, this radioactivity was considered a bonus. And promoted!

In the 1800s, J. J. Scheppe developed artificial mineral waters. They were very popular due to the poor quality of tap water available. Manufactured mineral waters (without radioactivity) are no longer sold as mineral waters. Legally, mineral waters are defined as waters from an underground source.

Contradictory gases

In Australia, the soft-drink manufacturing industry is worth over \$4 billion. Cans or bottles of Coca-Cola® and Pepsi® are probably found in most homes. These businesses were started by using the research undertaken by Joseph Priestley (1733–1804). Priestley is credited with discovering oxygen and his interest in gases was further advanced in 1767 when he moved next door to a brewery. He started a series of experiments with the brewery gas called ‘fixed air’.

Fixed air is carbon dioxide. Being non-polar, carbon dioxide is insoluble in water. Even though carbon dioxide is classed as insoluble, extremely low amounts can be present in solution. Carbon dioxide in water slowly acquires a shell of water molecules. This allows reactions to occur that result in the water becoming acidic. So despite the concentration of carbon dioxide being low, there is a detectable change in the water at the end, as shown by the reactions:



This slightly acidic taste is the pleasant taste that Priestley enjoyed. Soda water was invented. He shared his knowledge on how to make soda water. More flavourings were added to make the soft drinks we know today. Manufacturers use high pressures and very low temperatures to dissolve carbon dioxide at about 150 times the concentration in tap water.



To learn about how solubility is expressed, how to calculate the concentration of the solute and how to convert between different units (ppm and % v/v, % w/w and % v/w), refer to Chemistry section 6.7 on page 279.

The use of scientific knowledge may have beneficial and/or harmful and/or unintended consequences (ACSCH013 and ACSCH052)

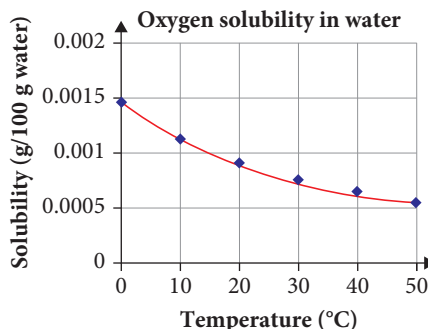
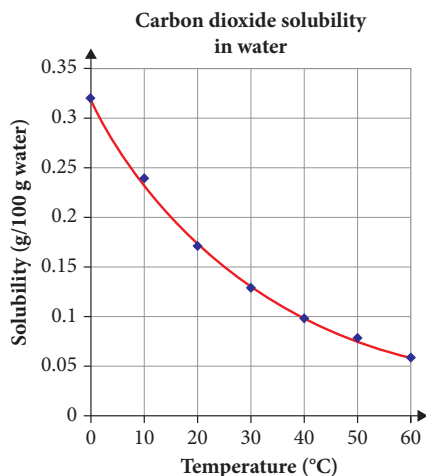


ACTIVITY 3.4

BREATHING AND OXYGEN

Land-based animals breathe oxygen in air, which then dissolves in the water that lines the lungs. Most aquatic organisms must obtain oxygen directly from the water.

Polar substances readily dissolve in water to give a relatively high concentration of solute. Concentrations of about 50 g in 100 g of water are not unusual. Non-polar substances do not readily dissolve; their solubility is usually measured in parts per million (ppm).



◀ **Figure 3.11**
Solubility of carbon dioxide and oxygen in water

- 1 Examine Figure 3.11.
 - a At what temperature is most oxygen dissolved in water?
 - b Compare the relative amounts of oxygen and carbon dioxide that can dissolve in water at a certain temperature. Propose a reason why this occurs.
 - c What is the relationship between temperature and oxygen solubility?
 - d Determine the amount of oxygen and carbon dioxide that dissolves in water at 37°C.
- 2 Colder regions of the ocean support more life than warmer regions. What do these facts indicate about the consequences of global warming heating the oceans?



To learn about the unit moles per litre that chemists use as the unit of concentration, refer to Chemistry section 7.2 on page 312.

Optimising performance for endurance sports

Water acts as a solvent in our bodies. Our body is about 70% water and each cell contains a lot of ions in solution. An important task of chemists is determining what substances are present in water and aqueous solutions and their concentrations. This is because the concentration of a substance is important.

The ions formed from the minerals sodium and potassium are important for transmitting electrical impulses along our nerves. Together with calcium ions, sodium and potassium ions are important for muscle contractions. Their salts are called **electrolytes** and a solution made by dissolving salts is called an electrolytic solution. Salt ions disperse when dissolved in water and the solution is then able to conduct electricity.

When you sweat, you lose these electrolytes, particularly potassium and sodium ions. It is important to maintain salt and water levels during exercise and sport. But it becomes more important when hot and when you are exercising for a long time. If you sweat and lose even 2% of your body weight, your blood volume decreases, which will cause your heart to pump harder. This reduces an athlete's performance.

When exercising for a long time, pure water is not enough to replace electrolytes. Sport drinks are designed to cater for this need. If they are labelled as **isotonic**, they claim to have the same salt, sugar or electrolyte level as the body.

Some drinks are **hypotonic** because the salt levels are lower than the body's salt levels. But if you consume these sport drinks when you are not exercising and are fully hydrated, you may be drinking a **hypertonic** solution. A hypertonic solution is one in which the salt level is higher than the surrounding fluid. This may actually make it harder for your body to absorb the water.

The difference between hypotonic, isotonic and hypertonic solutions is the concentration of solutes. You can dilute a hypertonic solution to make an isotonic solution.

▼ **Figure 3.12**
Isotonic, hypotonic and hypertonic solutions

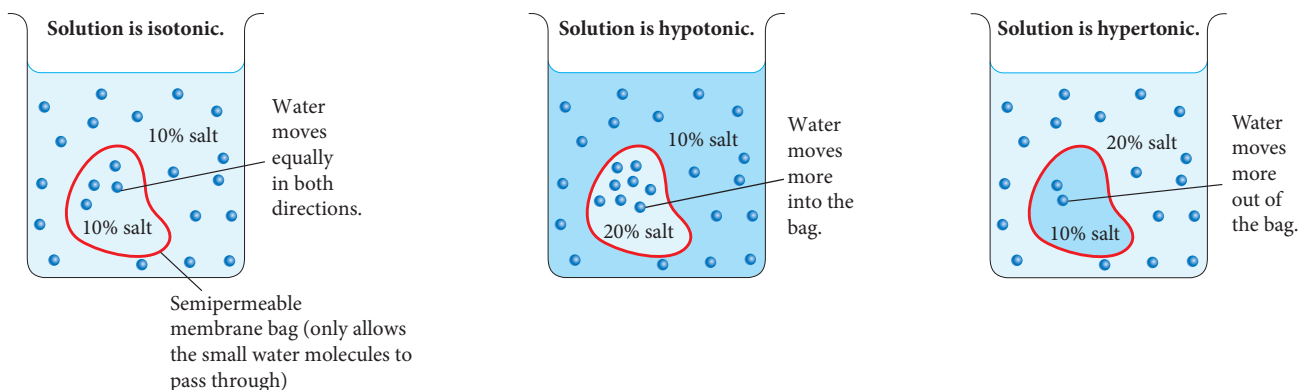




Photo by Brice Overham

◀ **Figure 3.13**

During endurance events it is important to maintain hydration and sugar levels. Cycling events such as the Murray to Moyne, in which teams ride 520 km over 2 days, are a challenge for most people. This cyclist, Samantha, has the added complication of having to also balance her own blood sugar levels due to her type-1 diabetes.

WOW

Water can be fatal

Drinking too much water too quickly can cause water intoxication and is potentially fatal. With too much water, the electrolytes in the body are diluted. Fatalities are rare and have occurred during water drinking contests, when trying to rehydrate after excessive exercise or when attempting to counteract the effects of drugs such as Ecstasy. In 17th- and 18th-century France, forcing people to drink water was a legal method of torture handed down by the French courts.

Scientific knowledge can be used to develop and evaluate projected economic, social and environmental impacts and to design action for sustainability (ACSCH015 and ACSCH054)



Blood chemistry

Blood is a complex **suspension** of plasma (fluid component) and cells. It is more viscous than water due to the dissolved salts and the biomolecules and cells being transported through the body. Polar substances can directly dissolve into the plasma. Non-polar substances such as many hormones have complex carrier proteins to transport them along the blood vessels. Blood is responsible for the delivery of oxygen and nutrients to all parts of the body, temperature regulation and waste removal.

Another role of blood is to control the pH of the body despite all the chemicals and reactions occurring. Blood has to maintain a pH of 7.4. There are three chemical buffer systems for this: the carbonate/carbonic acid buffer, the phosphate buffer and the buffering of plasma proteins. The carbonate (CO_3^{2-})/carbonic acid (H_2CO_3) buffer is the most important.



To learn about acids, bases, pH and their reactions, refer to Chemistry section 7.4 on page 320.

Blood monitoring

One vital nutrient transported by blood is sugar. The main blood sugar is glucose. Sugar levels are typically controlled between 79 and 110 mg per 100 mL of blood. In diabetes, the sugar levels in blood are not fully controlled by the body. People with diabetes have to inject a hormone (insulin) to decrease sugar levels. It can be fatal to have too low a level of sugar (hypoglycaemia). In the short term, low sugar levels are far more dangerous than high sugar levels (hyperglycaemia). The long-term health issues facing some people with diabetes are thought to be due to the high sugar levels. Blood becomes more viscous with excess sugar levels. This changes the microcirculation, particularly near the kidneys, liver and heart.

Diabetes Australia runs a national research program administered through the Diabetes Australia Research Trust (DART). Research is being undertaken in many areas from investigating different types of insulin to better monitoring and injection methods. Gone are the days when one injection of insulin was given and the person had to limit what they could eat. Now insulin can be long acting, short acting or designed for use during pregnancy. Insulin can be taken as a tablet, injected via a pen or continuously pumped in. This research is providing better health outcomes for people with diabetes.

There are many reasons why the level of particular chemicals in the blood is important. Using the known healthy range for particular substances, doctors can diagnose a health condition.



To learn more about how substances are detected by chromatography, refer to Chemistry section 6.9 on page 289.

ACTIVITY 3.5

BLUE BLOOD

Oxygen-rich blood is red. Blue blood indicates deoxygenated blood. When deprived of oxygen, cells will die. Permanent damage may occur within 3 minutes.

Use Figure 3.11 (page 71) to determine the amount of oxygen that can be dissolved at 37°C. This level is too low to sustain life. Unlike carbon dioxide, oxygen does not react with water, so it is transported around the body by a molecule called haemoglobin.

Conduct an Internet search to determine how haemoglobin works and then complete one of the following tasks.

- Compare haemoglobin to other respiratory pigments, such as myoglobin or foetal haemoglobin.
- Compare haemoglobin to the range of respiratory pigments used by some invertebrates, the true 'blue bloods'.

Copper – the Jekyll and Hyde element

Copper is an essential trace element. The recommended minimum level dietary intake for infants is around 200 µg/day. Copper is toxic, so maximum levels in drinking water are around 2 ppm. Determining the copper levels in blood can indicate if too much or too little copper is present for good health.

In Chapter 4, you learnt about atomic absorption spectroscopy (AAS). The first few atomic absorption spectrometers built by Alan Walsh in 1954 were used to solve medical treatment problems. The value of this technique can be seen when sick infants are suspected of having copper poisoning. Infants do not have much blood, so a technique that only requires a small amount of blood for analysis is very useful. AAS can confirm, with minimal harm, high copper levels in an infant's blood.

Hot water systems are typically made of copper. Copper dissolves more readily in hot water than in cold water. If baby formula is made up with hot water, it can contain high enough levels of copper ions to cause copper poisoning and even death. The infant's small size and their reliance on baby formula as their sole source of nutrients makes this an important issue.

You should not use hot water for cooking or drinking. Always let the cold water run for 30–60 seconds, particularly if you are preparing baby formula.

Advances in science understanding in one field can influence other areas of science, technology and engineering (ACSCH011 and ACSCH050)



To revise atomic absorption spectroscopy, refer to Chemistry section 1.9 on page 127.

What is in your water?

Australia is the driest continent, so the supply of water is an important issue. In a drought region, sources of water such as bore water may be investigated. In other areas after a flood, the quality of the water can be cause for concern. We need to have a clean pure source of water for our health, but not all water sources are pure. Scientists use different methods of analysing water depending on what they are looking for; for example, salt, organic compounds or gases. Some of these methods are listed in Table 3.3.

Table 3.3 The different solutes present in water require different techniques to detect them

Solute	Technique	Key points	Relevant chapter
Salts	Range of precipitation reactions	Need to know solubility rules and common precipitates	Chemistry Chapter 7
	AAS	Metal ions detected to ppm	Chemistry Chapter 1
Organic compounds	Chromatography	Relies on the relative solubility of the different substances	Chemistry Chapter 6
Gases	Gas chromatography	Relies on the relative solubility of the different substances	Chemistry Chapter 6

Heavy metals

One of the main concerns with water supplies is the presence of heavy metals. These are metal or metalloid elements that are at least five times denser than water. The most common heavy metals of concern are arsenic, cadmium, iron, lead and mercury.

Heavy metals can enter the body in food, water, or air or by absorption through the skin. Some heavy metals such as copper, iron and zinc are essential to life at low levels and can be found in many food items. Heavy metals become toxic when they accumulate in the soft tissues of the body. Acute poisoning normally occurs after inhalation. The presence of heavy metals in water can slowly accumulate over time to dangerous levels, causing major health issues.

Ouch ouch disease

The first documented case of mass cadmium poisoning occurred in Japan. Itai-itai (ouch ouch) disease was first diagnosed in the 1950s although it had been noticed in 1912. This disease is named after the painful cry of the patients. Cadmium is of no benefit to the body. Even at low levels of exposure, cadmium will accumulate in and poison soft tissues such as the kidneys and liver. At higher levels, the kidneys start to fail and cannot remove acid from the blood. The side effect of this is the development of gout, a painful condition. High levels of cadmium cause bones to soften and fractures become more common.

In Japan, the exposure to cadmium was due to mining along the river that was the source of water for the rice fields. Although mining for other metals had occurred along the river for centuries, increased demand during World Wars I and II caused the outbreaks. The worst symptoms occurred in mothers who had borne many children. Cadmium enters the body in the same way as zinc. In pregnancy, there is an increase demand for zinc. The body absorbs more zinc (which means correspondingly higher levels of cadmium).

Science is a global enterprise that relies on clear communication, international conventions, peer review and reproducibility (ACSCH009 and ACSCH048)



To learn about how precipitation reactions can be used to detect metal ions in solution, refer to Chemistry section 7.1 on page 304.



To learn how dilutions affect concentration, refer to Chemistry section 7.3 on page 317.

ACTIVITY 3.6

WHAT TO DO?

You work for the EPA (Environmental Protection Authority) and have just been advised that water contaminated with cadmium has been released into a local lake.

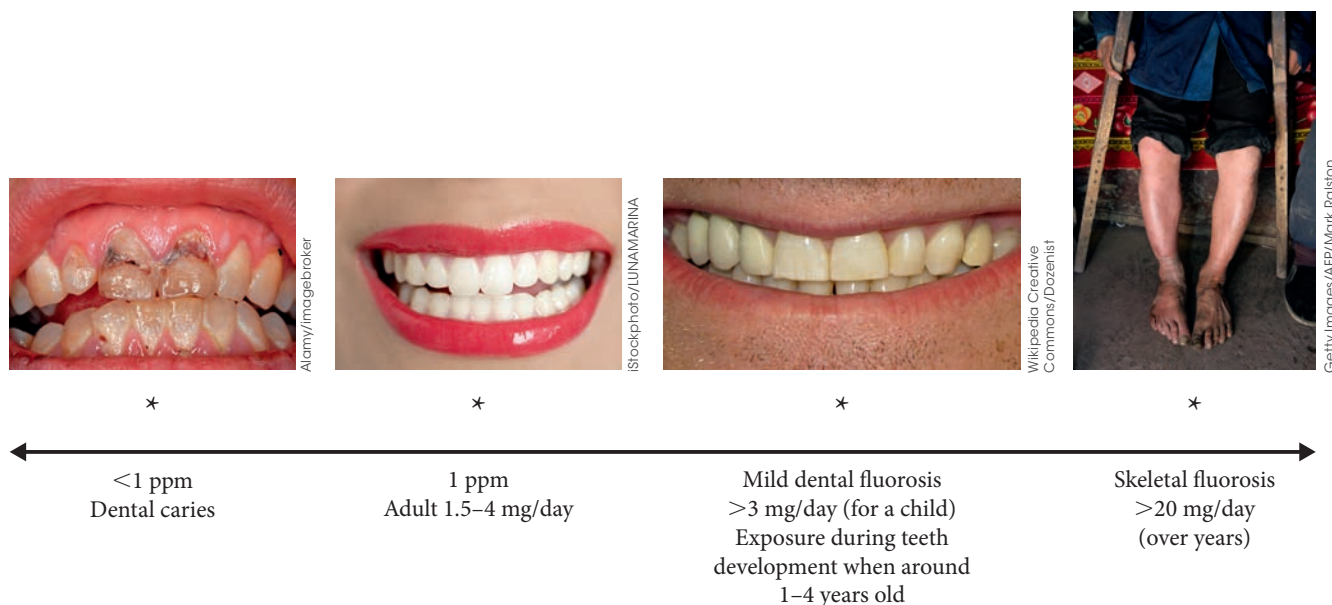
A sample of the industrial wastewater contains $3000 \mu\text{g g}^{-1}$ cadmium ions and every 100 L of this waste is diluted with 100 L of pure water before it is released.

- What is the concentration of cadmium ions in the new, diluted, solution?
- A total of 500 L of this dilute solution was released into a lake that contains 1×10^6 L of water. If there was no cadmium in the lake initially, what approximate concentration would it have after the solution was added?
- The cadmium concentration in the unpolluted water is generally less than $1 \mu\text{g L}^{-1}$. How does this compare to the lake after diluted wastewater containing cadmium is added?
- The World Health Organization has reported the cadmium uptake by humans in a smoker's house to be 2–4 μg per day. How many glasses (300 mL) of the lake water would you need to drink per day to reach this amount?
- What would your advice be to the town council, the manager of the site? Would you drink the water and would it be any different if you were pregnant?

Fluoridated water

Fluoride, in the form of sodium fluoride (NaF), has been added to the water of most Australian states for decades to reduce tooth decay. Fluoride has been added at a level of 1 ppm. This is similar to the level in seawater, while lake water has typical levels around 0.5 ppm. At these low concentrations, fluoride significantly reduces the number of dental cavities. However, when the natural level in the water is already high, dental fluorosis (discoloration or staining of teeth) can occur. With excessive exposure to fluoride, a more severe and painful disease of the bones and joints, skeletal fluorosis, may occur. Excessive fluoride levels are generally due to industrial pollution, very high levels in local water or volcanic activity. It can occur when the water supply originates from certain rocks in parts of India, Africa and the Middle East. In some areas, most of the water supplies contain above the recommended levels, sometimes as high as 45 mg L^{-1} . Some bores in Kenya have values between 1640 and 2880 mg L^{-1} .

While some people have expressed concern about the fluoridation of their water, others are concerned about the increased consumption of bottled water as it does not have the recommended levels of fluoride.



▲ **Figure 3.14**

The relevance of fluoride in water to health was first noticed in native-born residents of Colorado Springs, Colorado, USA. Cosmetic stains on teeth were found to be caused by high levels of fluoride in the local water supply. People with these stains also had an unusually high resistance to dental cavities.

ACTIVITY 3.7

BABY TEETH AND GOOD HABITS

Children's toothpaste has a lower amount of fluoride ions than adult toothpaste.

- Explain why fluoride is added to most toothpastes.
- Explain why children's toothpaste has lower levels of fluoride ions.
- What other measure would you recommend for children?

Managing water quality

Australia has guidelines to ensure the quality of the drinking water. These are determined in collaboration with National Health and Medical Research Council and the Natural Resource Management Ministerial Council. These guidelines help manage the water supplies and protect the safety of the community.

Scientists are constantly developing more accurate and reliable measurements of the substances in the waterways. Metals are detected at very low levels by AAS. Sometimes, chemists can use knowledge about precipitation reactions to work out what is present.

Detecting organics in water

Coal and coal seam gas (CSG) is natural gas (methane) and is found in coal deposits. CSG is formed from plant matter that has been under extremely high pressure for millions of years. CSG is used in the same way as any other form of natural gas, such as for cooking or generating electricity. CSG is plentiful in Australia.

CSG is extracted by drilling a well into a coal seam and pumping in large volumes of water and sand at high pressure. This fractures the coal seam and releases the gas. The sand deposits in the fractures to prevent them closing when pumping pressure ceases.

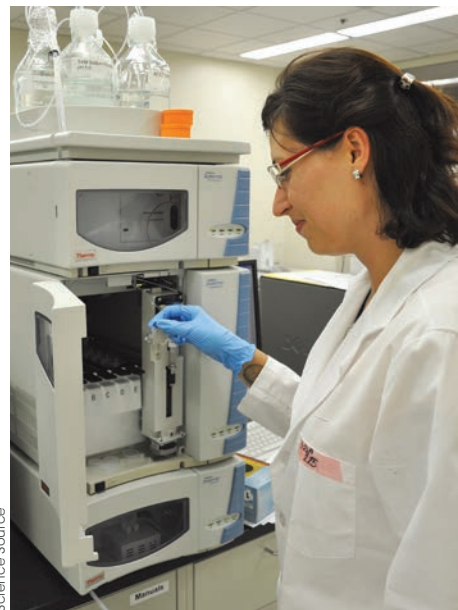
CSG can have environmental and economic benefits in a low-carbon future. Not everyone agrees that this industry should develop. The debate over CSG has united farmers and environmentalists. Environmentalists are concerned about damage to the landscape while rural community members may also see a threat to their way of life.

One concern is the amount of water that is used to extract the methane. There is a risk that the contamination of aquifers may occur because of:

- chemicals injected through the fracking process (in which rocks are fractured by injecting chemicals under high pressure)
- chemicals released from the coal seam
- removed water being replaced by water flowing in from other aquifers into the coal seam.



Alamy/Mikael Karlsson



Science Source

▲ Figure 3.15

Chromatography instruments: gas chromatography and high-performance liquid chromatography are analytical techniques used to separate organic substances.

Science is a global enterprise that relies on clear communication, international conventions, peer review and reproducibility (ACSCH009 and ACSCH048)



DRINKING WATER GUIDELINES

Visit this website to read the latest drinking water guidelines in Australia.



To learn about how to identify unknown ions in solution, refer to Chemistry section 7.7 on page 332.



COAL SEAM GAS

Visit this website to learn more about the issues associated with coal seam gas.

Scientific knowledge can be used to develop and evaluate projected economic, social and environmental impacts and to design action for sustainability (ACSCH015 and ACSCH054)



To learn about chromatography techniques, refer to Chemistry section 6.9 on page 289.

Sensitive techniques such as gas chromatography and high-performance liquid chromatography can be used to detect very small amounts of possible pollutants. These techniques rely on surface interactions to separate substances on the basis of their relative differences in attraction between a solid and a mobile phase.



SAFE WATER SUPPLIES

Water treatments are vital for securing safe water supplies. To learn how it is done, take one of Seqwater's (Queensland) virtual tours.

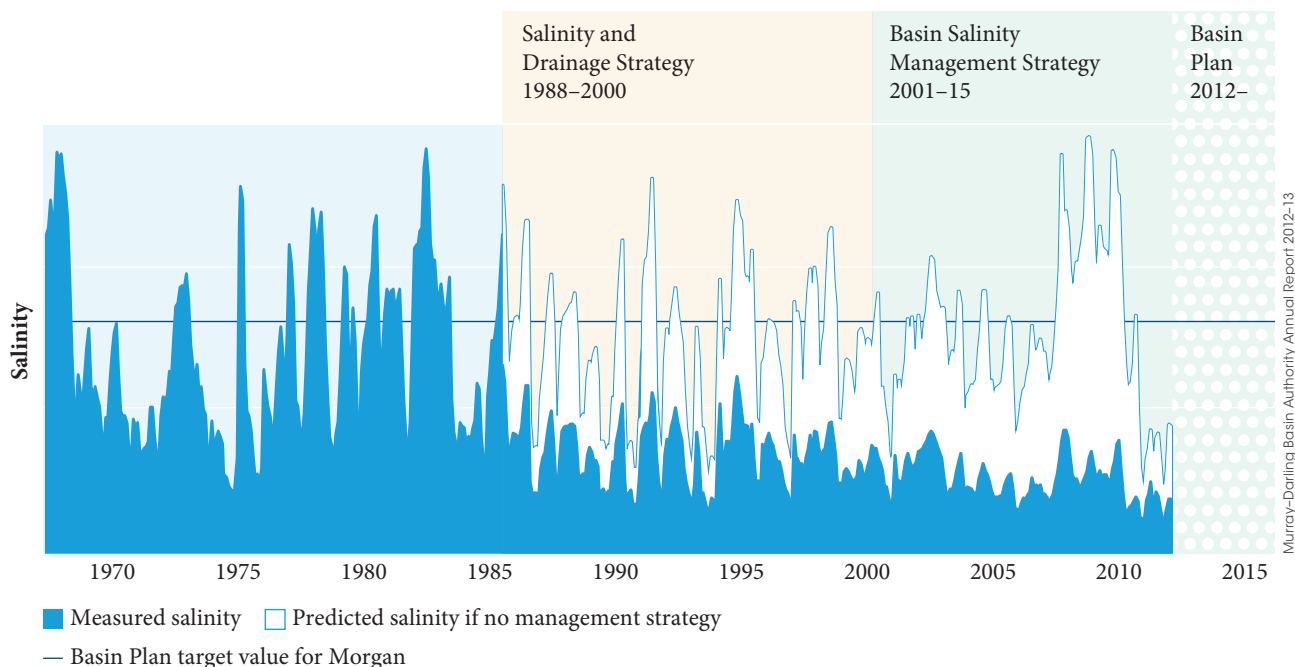
Water in crisis?

Australia is a dry continent with a growing population. This makes water availability an important sustainability issue. Rivers, lakes and ocean may be vastly different but the threats to them are remarkably similar. Some water supply issues are due to the European settlers clearing the land and not understanding Australia's unique environment. Climate change is adding another dimension to the problem. The availability of reliable fresh supplies of water is something that governments have to consider. Many cities around Australia have water treatment plants that remove the salts from seawater or highly saline water to produce drinking water. In Australia, a variety of techniques are used to produce the energy required for these processes, such as wind farms.

ACTIVITY 3.8

SALINITY

- 1 Salinity is a large problem in particular regions of Australia. The Murray River runs along the border of Victoria and New South Wales. It is the source of Adelaide's water.
 - a Find out two main sources of the salt in the Murray River.
 - b What could cause the variation in salinity observed in Figure 3.16?



▲ **Figure 3.16**

The effects of salinity management in the Murray–Darling Basin: average salinity levels in the Murray River at Morgan, South Australia, 1970–2010

- c Why is it important to examine water systems over time?
 - d Name some of the interventions that have been implemented to decrease salinity levels.
 - e What would be some consequences of not intervening?
 - f Adelaide is down river from Morgan. Explain how the graph could have changed if the testing was done in Adelaide.
 - g What other options could be available to the cities to guarantee water supply?
 - h Besides the supply of water to towns and cities, what problems arise from high salinity in the rivers?
- 2 Table 3.5 lists typical calcium carbonate levels of various waters around Australia. Calcium carbonate results in **hard water**, which is water that soap won't lather in and leaves lime scale in kettles.
- a Which city has the softest water?
 - b Which has the hardest water?
 - c Suggest why the composition of water varies from city to city.

Table 3.5 Calcium carbonate levels in potable (drinkable) water in major cities in Australia

Major city	CaCO ₃ concentration (ppm)
Adelaide	134–148
Brisbane	100
Canberra	40
Darwin	31
Hobart	5.8–34.4
Melbourne	10–26
Perth	29–226
Sydney	39.4–60.1

Extension

In Singapore, drinking water is produced from recycled sewage. Research to find out why we don't recycle water in Australia.

Great Barrier Reef

The Great Barrier Reef is the world's largest reef and it is visible from outer space (Figure 3.17). This area of more than 900 islands stretches for more than 2600 km. It has been on the UNESCO World Heritage List since 1981. The reef has a very high biodiversity, being home to the dugong, six species of turtles, more than 300 species of hard, reef-building corals, 400 species of sponges, 240 species of birds and more than 1500 species of fish.

Major developments along the coast of Queensland, such as farming, aquaculture and urban development, have increased the run-off to the reef. Stormwater is water that runs off after rain falls in an urban area and then washes down to the ocean. It carries with it the oil, petrol, dog droppings, litter and sediments that were on the streets. Rain in rural areas may include livestock waste, fertilisers and pesticides. Approximately 25% of the land drains directly into the reef. This run off affects the **turbidity** of the water. Turbidity is a measure of how much the suspended particles reduce the amount of light that penetrates the water. Increased turbidity reduces photosynthesis and blocks fish's gills.

Run-off from the land consists of soluble fertiliser ions such as nitrate (NO₃⁻) and phosphate (PO₄³⁻). These ions are used by the single-celled, plant-like organisms called **phytoplankton**;

Scientific knowledge can be used to develop and evaluate projected economic, social and environmental impacts and to design action for sustainability (ACSCH015 and ACSCH054)



Figure 3.17 ►
The Great Barrier Reef is so large that it is visible from space.



Getty Images/Universal Images Group

an increased supply of nitrates and phosphates can cause algae blooms or **eutrophication**. Algal blooms block the light to the organisms such as corals or the sea grass beds in the Hinchinbrook Channel near Townsville. This reduces the food source for some vulnerable species such as the dugong and green turtles.

Run-off increases the amount of organic matter and heavy metals entering the waters. Microbes will feed on this matter, consuming oxygen in the water. There is an increase in the biochemical oxygen demand (BOD) in the water, which deprives other aquatic organisms such as fish of oxygen. Mining in the area has led to increased levels of cadmium, mercury, copper, lead, nickel, zinc and arsenic. Near the Great Barrier Reef catchment area, coal and shale oil are mined north of Gladstone, silica is mined near Cape Flattery and magnesia is mined north of Rockhampton. With increased mining, there is always a demand for bigger ports to bring the ships in and out of the area.

Ocean acidification due to high levels of carbon dioxide is a major threat. Higher acidity levels mean less carbonate is available for the coral and they are less able to build their skeletons and repair themselves when damaged. A recent study showed a 14% decrease in coral growth since 1990 due to increased levels of carbon dioxide. This was the most significant decrease in coral growth in the last 400 years.

The value of the Great Barrier Reef was estimated to be over \$51 billion by the Oxford Economic Group in 2010. The agricultural production is estimated to be worth \$2.3 billion per year, fishing \$500 million, tourism \$4.3 billion and recreation \$500 million. What would you spend to save it?

CHAPTER SUMMARY

- Water has unique properties that are vital to the biological, chemical and physical processes on Earth. Water exists naturally in three states: solid (ice), liquid (oceans, lakes and rivers) and gas (water vapour).
- Strong hydrogen bonding in water accounts for all the properties that relate to temperature. A relatively large amount of energy is required to break the bonds between the molecules of water to change state from liquid to gas (high latent heat) or to increase the average temperature of the water (high specific heat).
- Evaporation of water has a cooling effect: bodies of water have less extreme temperature ranges than land; and water is liquid over a large temperature range. These phenomena enable organisms to survive on Earth.
- The cohesive nature of the water molecules explains surface tension. Understanding this allowed techniques such as chromatography to be developed and the use of surfactants in the medical, detergent and food industries.
- Ice floats because of the regular crystal structure adopted upon freezing. This protects the life under the ice in the lake from freezing and alters the density of ocean waters.
- Water is the universal solvent, partly because it is liquid from 0°C to 100°C. It can readily dissolve polar substances, such as salts and polar molecules. Salt levels change the temperature at which water freezes, allowing some cells to survive extreme temperatures. Salinity affects the density of the water, which, in turn, affects the convection currents of the ocean.
- Sometimes, there are detrimental effects if the concentration of solutes is too low or too high. This knowledge is used in medical diagnosis and in determining water quality. Techniques such as chromatography can determine the amounts of solute present.
- Understanding water chemistry and quality enables the environmental, social and economic impact of human activity to be developed and evaluated, such as how higher carbon dioxide levels affect the acidification of ocean ecosystems, which in turn affect industries such as fishing and ecotourism.

CHAPTER GLOSSARY

chromatography techniques used to separate components of aqueous, liquid or gaseous mixtures

cyroprotectant a chemical that is used to stop biological tissues from freezing

El Niño a warm ocean current that develops every few years along the coast of Ecuador and Peru

electrolyte a charged solute that allows the solution to conduct electricity

enzyme a protein molecule that catalyses a specific type of reaction by lowering the energy of activation

eutrophication a process in which an additional amount of nutrients leads to excessive growth of plants including microscopic plant-like phytoplankton

hard water water that contains high levels of Ca^{2+} and Mg^{2+} ions, which interferes with the action of soaps

hydrophilic 'water-loving', a particle with polar regions that bond to water

hydrophobic 'water-hating', a particle with mostly non-polar regions that do not bond with water

hypertonic concentration of solutes is higher than another solution

hypotonic concentration of solutes is lower than another solution

isotonic concentration of solutes is the same in both solutions

ocean acidification lowering of pH of the ocean due to the reaction of carbon dioxide with water molecules

phytoplankton microscopic plant-like organisms found at the surface of oceans, seas and lakes

solute the substance that is dissolved or the smaller component of a solution

solution a homogeneous mixture that is formed when a solute dissolves in a solvent

solvent the substance in which the solute dissolves or the greater part of a solution

supersaturated solution an unstable solution that has more solute than possible at that temperature; it is formed from dissolving solute into the solution at a higher temperature and then allowing it to cool. The extra solute is still in solution but can readily be crystallised

surfactant a chemical that lowers the surface tension of a liquid

surface tension the force that arises from the attraction of the surface molecules to the bulk of the material

suspension a cloudy, heterogeneous mixture containing solid particles that will eventually settle out

turbidity cloudiness of water due to suspended microscopic particles. High turbidity reduces how far light penetrates into water

viscosity the ability of a fluid to resist flow; honey has high viscosity, water has low viscosity

water of crystallisation the water molecules that are bonded inside crystals; there is an exact ratio between the water molecules and either the salt or the polar molecule because of the number of ion-dipole or hydrogen bonds present between the two